

Aromatic Substitution Reactions

Organic Chemistry II

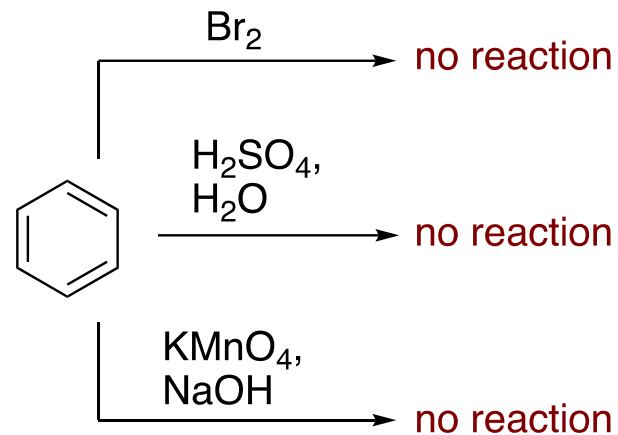
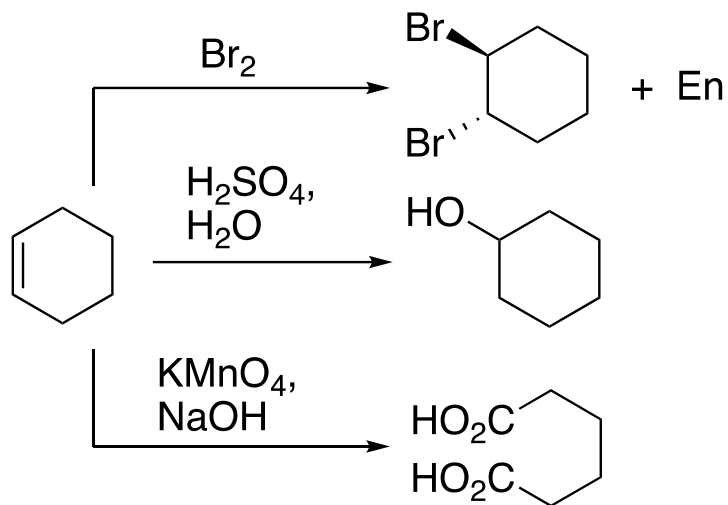
CHEM 2312

Dr. Michael Evans

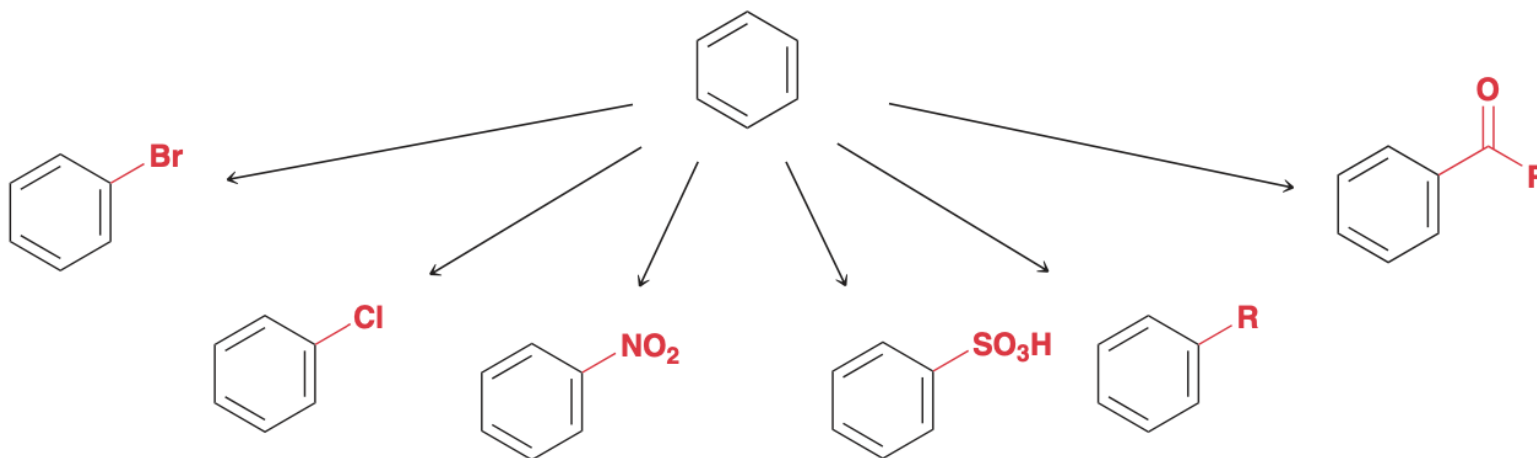
1. Describe the general mechanism of electrophilic aromatic substitution (EAS) using elementary steps and a reaction coordinate diagram.
2. Predict the products of simple EAS reactions of benzenes.
3. Predict the products of EAS reactions of substituted benzenes.
4. Predict and rationalize the relative rates of reaction of substituted benzenes in EAS reactions.
5. Apply functional group interchange reactions of benzene side chains in synthesis, including reductions of aryl ketones, benzylic oxidation, benzylic bromination, and reduction of nitro groups.
6. Plan multi-step syntheses of substituted benzenes using EAS reactions as key steps.
7. Describe the addition-elimination mechanism of nucleophilic aromatic substitution (NAS) using elementary steps and a reaction coordinate diagram.
8. Predict the products of NAS reactions of electron-poor benzenes and heteroarenes.
9. Describe the elimination addition mechanism of NAS using elementary steps; predict the products of reactions involving benzyne intermediates.

18.1 Aromatic Compounds Undergo Substitution, Not Addition

- Benzene is unreactive in typical reactions of alkenes



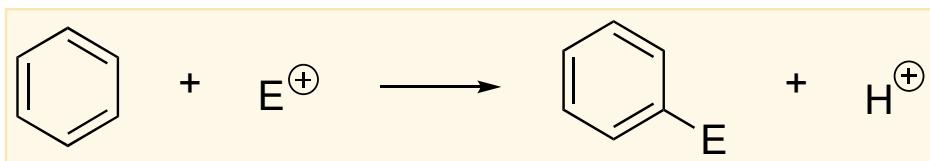
- However, benzene undergoes substitution in the presence of strong E^+ (**electrophilic aromatic substitution**)



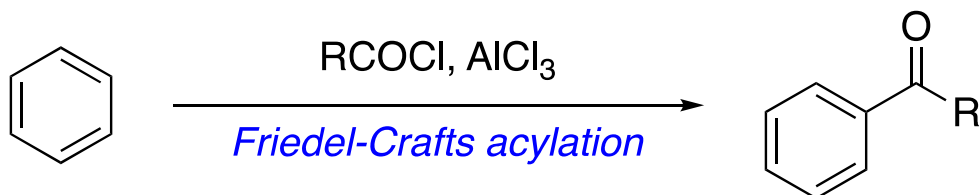
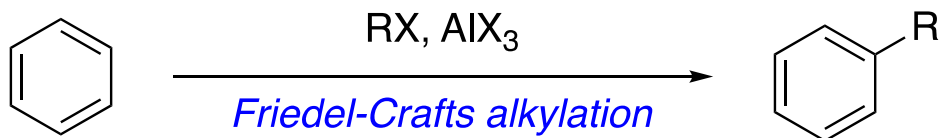
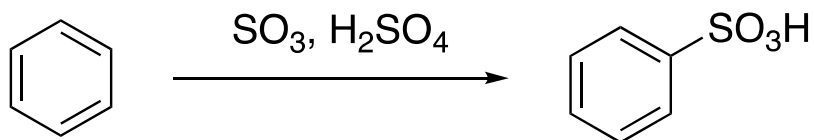
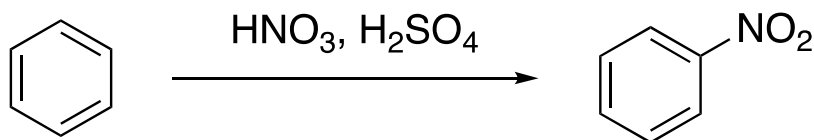
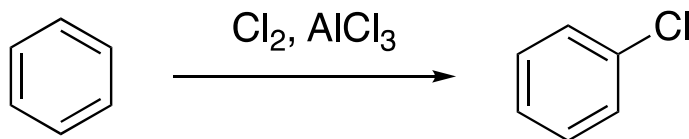
18.1 Electrophilic Aromatic Substitution | 1 of 2

- In an **electrophilic aromatic substitution (EAS) reaction** an electrophile E^+ substitutes for H^+
- Different reagents for EAS lead to the generation of unique electrophiles E^+

Active Electrophile (E^+)

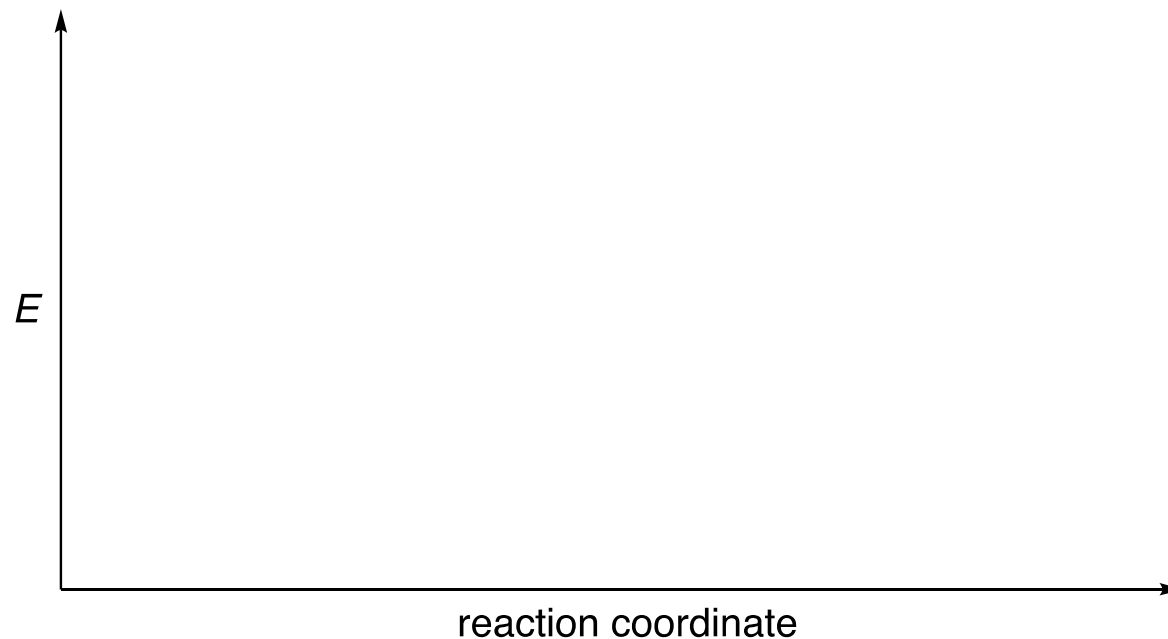
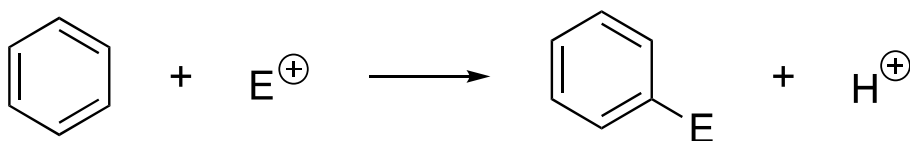


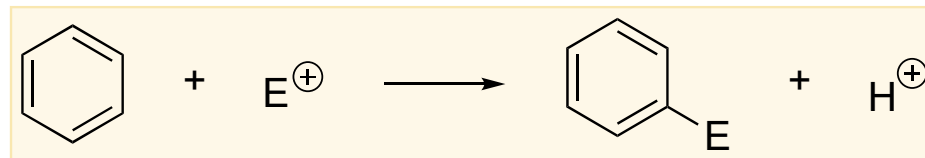
- What is the nature of E^+ and how is it generated?
- What are the relative rates of reaction of electron-rich and electron-poor benzenes?
- What is the regioselectivity in reactions of substituted benzenes?
- How do heteroarenes react?



18.1 Electrophilic Aromatic Substitution | 2 of 2

- The general mechanism of EAS involves coordination of the electrophile to the aromatic ring (A_E) followed by loss of a proton (pt)
- The intermediate cation is called an **arenium ion** or less commonly, a **sigma complex**
- The intermediate is non-aromatic! Addition of the electrophile is the slow step





Halogenation (X_2 , FeX_3)

Nitration (HNO_3 , H_2SO_4)

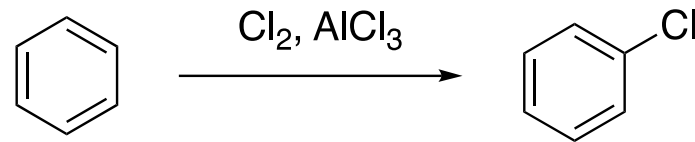
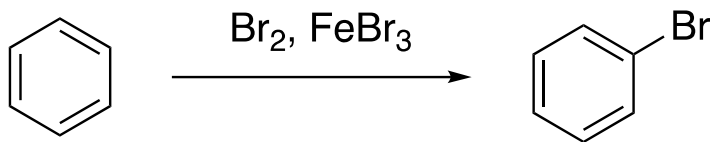
Sulfonation (SO_3 , H_2SO_4)

Friedel-Crafts alkylation (RX , AlX_3)

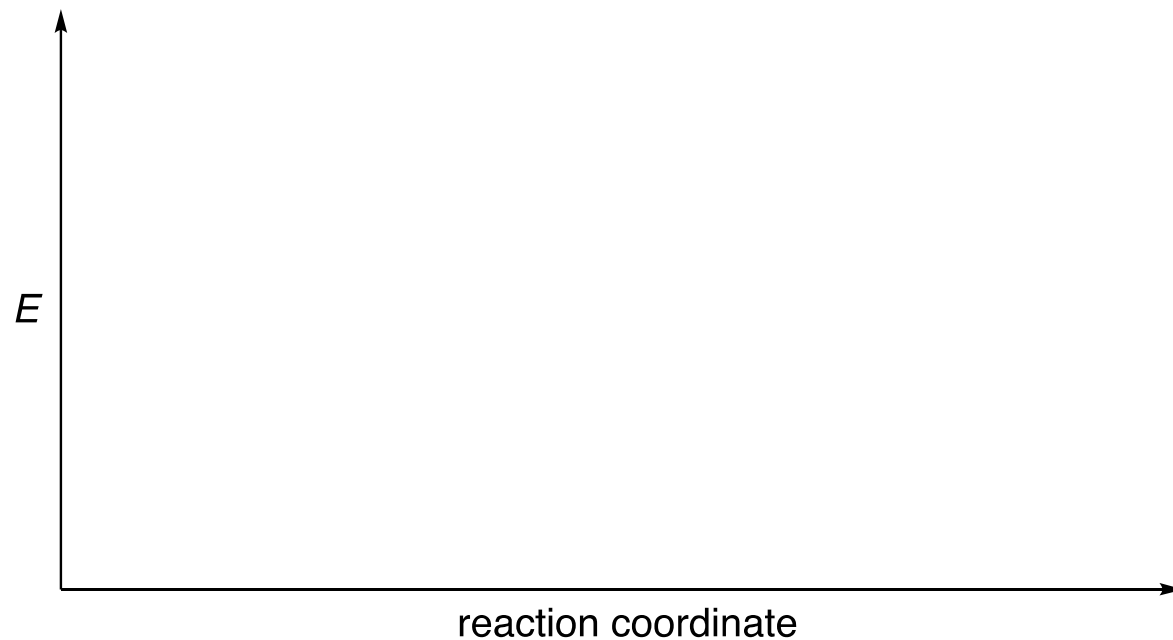
Friedel-Crafts acylation ($RCOX$, AlX_3)

18.2 Halogenation

- Elemental halogen (Cl_2 , Br_2) is combined with a Lewis acid (FeBr_3 , AlBr_3 , AlCl_3)

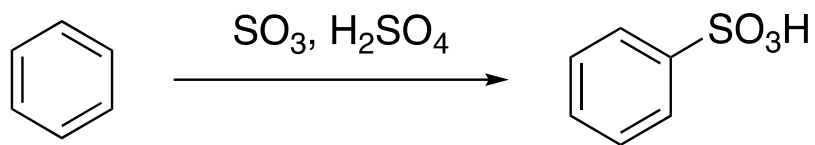


- A halogen atom coordinates to the Lewis acidic metal center (A_N)
- Benzene attacks the terminal halogen ($[\text{A}_\text{E} + \text{S}_\text{N}2]$), beginning the classical mechanism of EAS

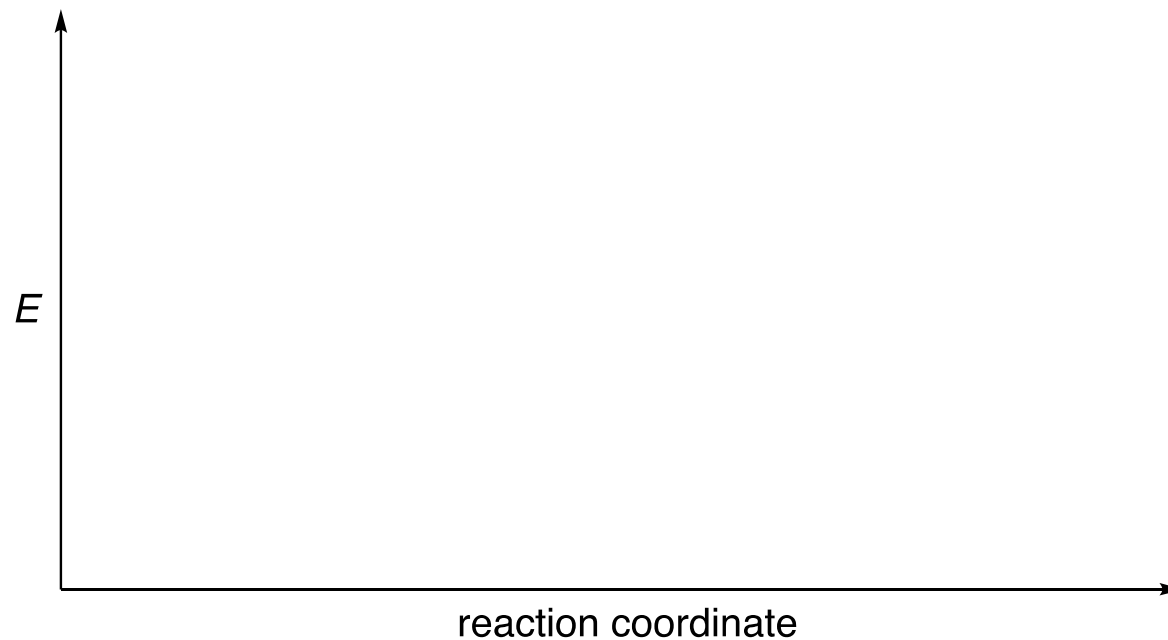


18.3 Sulfonation

- Sulfur trioxide (SO_3) is combined with concentrated sulfuric acid (H_2SO_4)

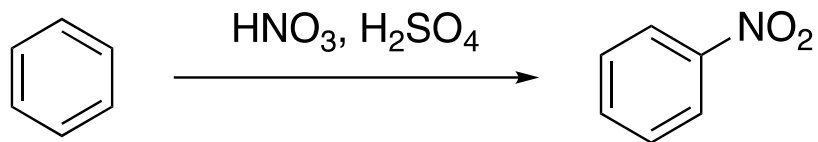


- Sulfuric acid protonates sulfur trioxide (pt), yielding the active electrophile SO_3H^+
- Benzene attacks the electrophilic sulfur atom (A_E), beginning the classical mechanism of EAS

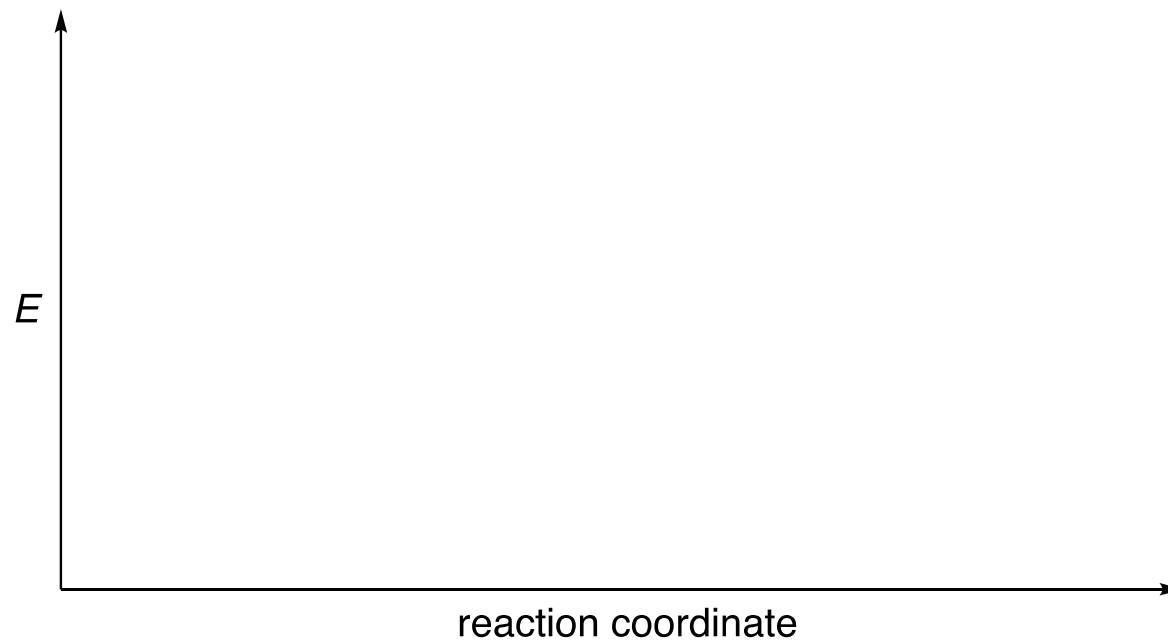


18.4 Nitration

- Nitric acid (HNO_3) is combined with concentrated sulfuric acid (H_2SO_4)

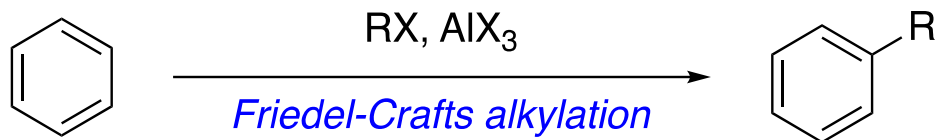


- Sulfuric acid protonates nitric acid ($\text{p}t$) and water is subsequently lost (E_β), yielding a nitronium cation (NO_2^+)
- Benzene attacks the electrophilic nitrogen atom ($[\text{A}_\text{E} + \text{Ad}_\text{N}]$), beginning the classical mechanism of EAS



18.5 Friedel-Crafts Alkylation

- An alkyl chloride or bromide (RX) is combined with a Lewis acid (AlCl_3)

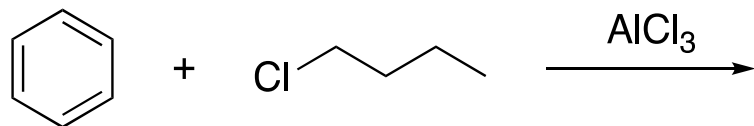


- As in halogenation, the halogen in RX coordinates to the Lewis acidic metal center (A_N)
- The XAlCl_3^- anion departs as a leaving group (D_N), yielding a carbocation R^+
- Benzene attacks the carbocation (A_E), beginning the classical mechanism of EAS

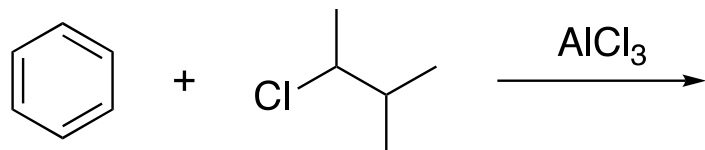
- Primary alkyl halides react with the arene via $\text{S}_\text{N}2$

18.5 Beware 1,2-Rearrangements in Friedel-Crafts Alkylation

- Reaction of *n*-butyl chloride with benzene in AlCl_3 does not give exclusively *n*-butylbenzene

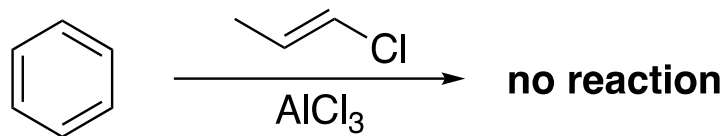


- Secondary carbocations will rearrange to tertiary carbocations via 1,2R when possible

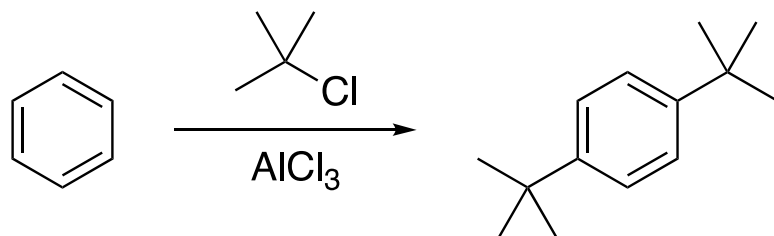


18.5 Limitations of Friedel-Crafts Alkylation

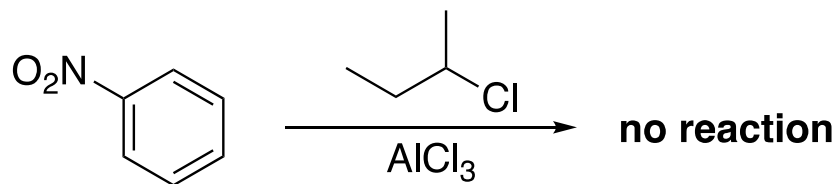
- The organohalide *must* be an alkyl halide with a C(sp³)-X bond



- Because alkyl groups are electron donating, alkylation activates the benzene ring toward further EAS reaction

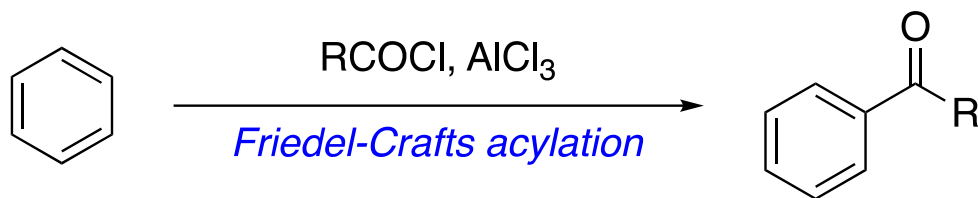


- Severely electron-poor arenes, such as nitrobenzenes, react at unacceptably slow rates



18.6 Friedel-Crafts Acylation

- An acyl chloride (ROCl) is combined with a Lewis acid (AlCl_3)

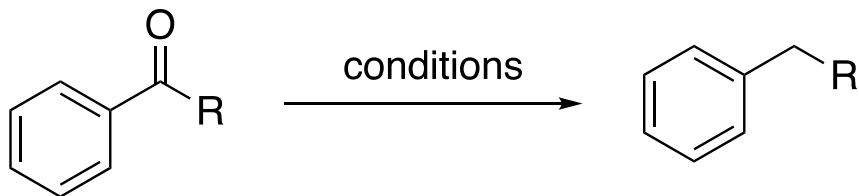


- As in halogenation, the halogen in RX coordinates to the Lewis acidic metal center (A_N)
- The XAlCl_3^- anion departs as a leaving group (D_N), yielding an **acylium ion RCO^+**
- Benzene attacks the acylium ion ($[\text{A}_\text{E} + \text{Ad}_\text{N}]$), beginning the classical mechanism of EAS

- Acylium ions do not rearrange and polyacylation is not an issue!

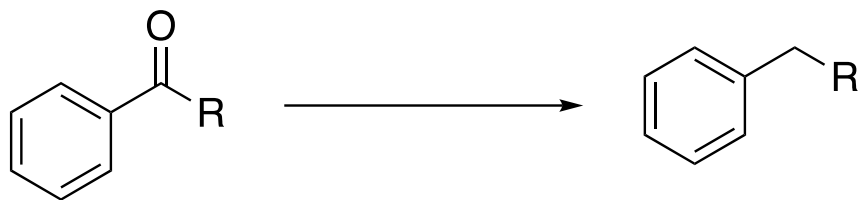
18.6 Reductions of Aryl Ketones

- Reduction of aryl ketones is combined with Friedel-Crafts acylation to synthesize primary arylalkanes ArCH_2R

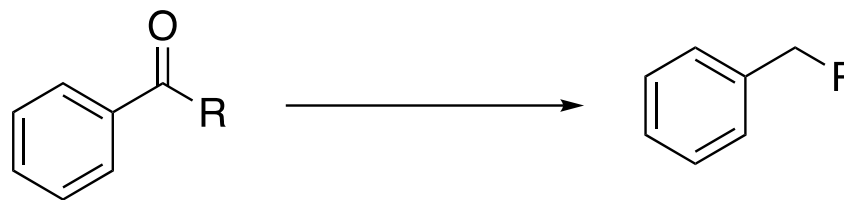


- This reduction can be effected under acidic or basic conditions

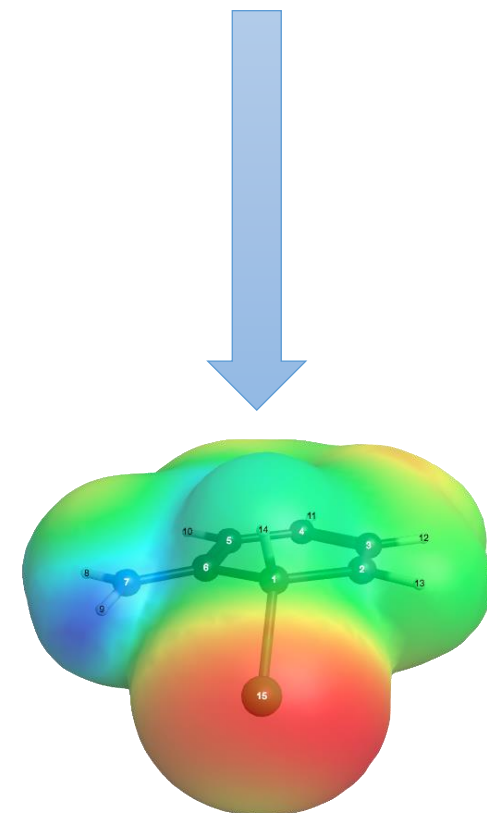
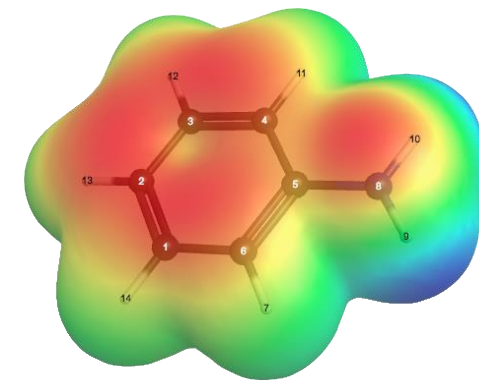
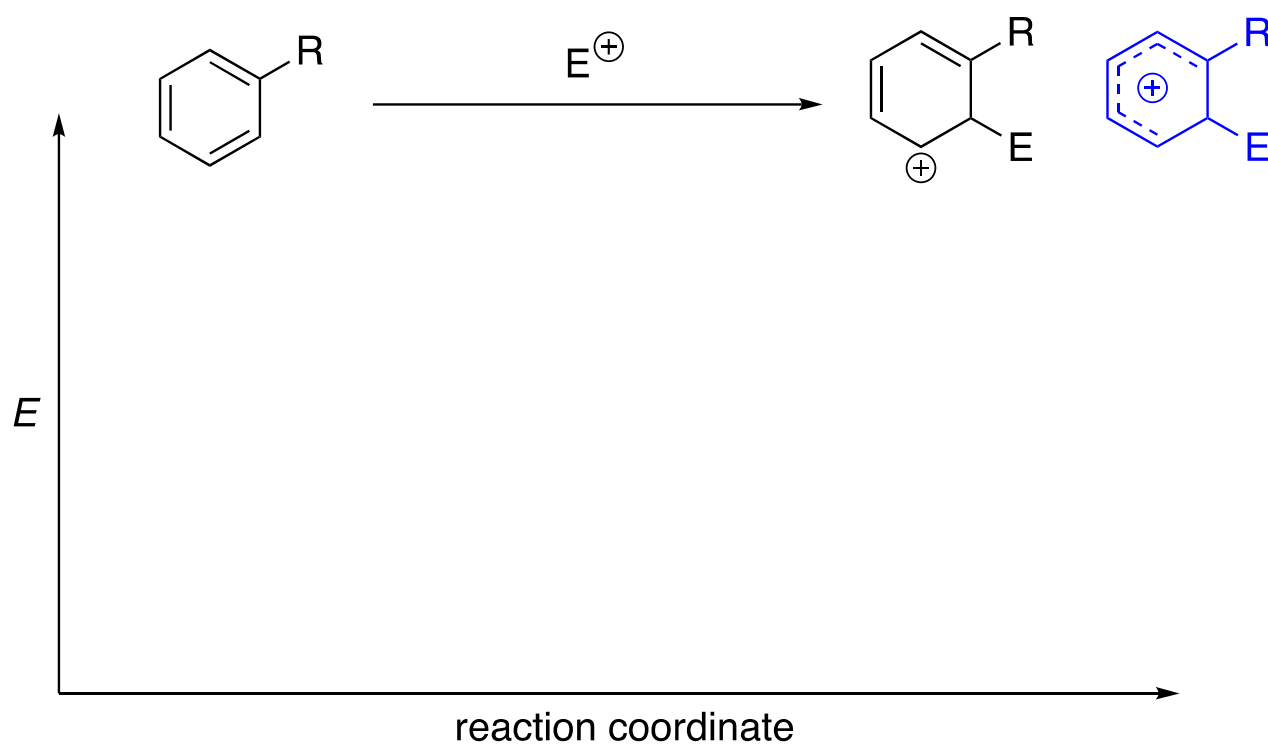
Wolff-Kishner reduction



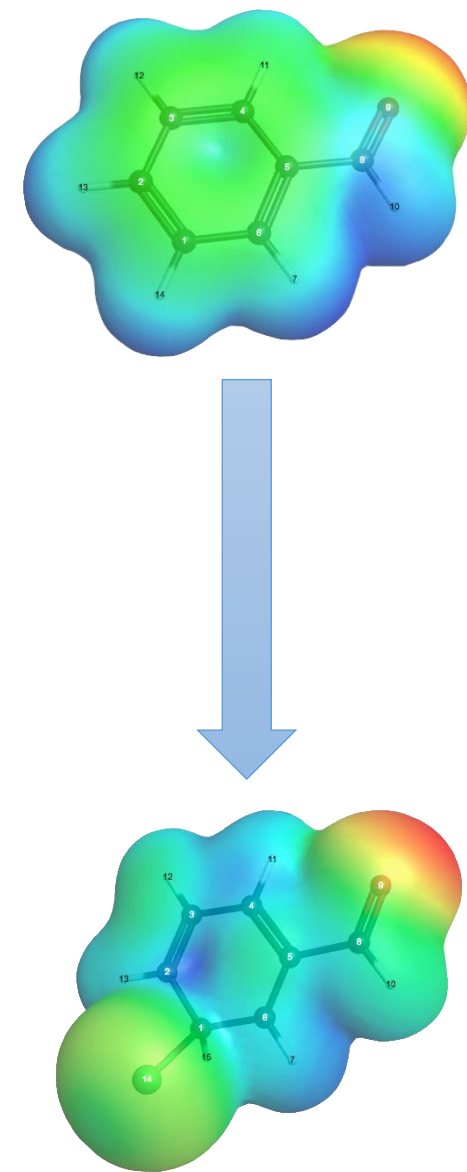
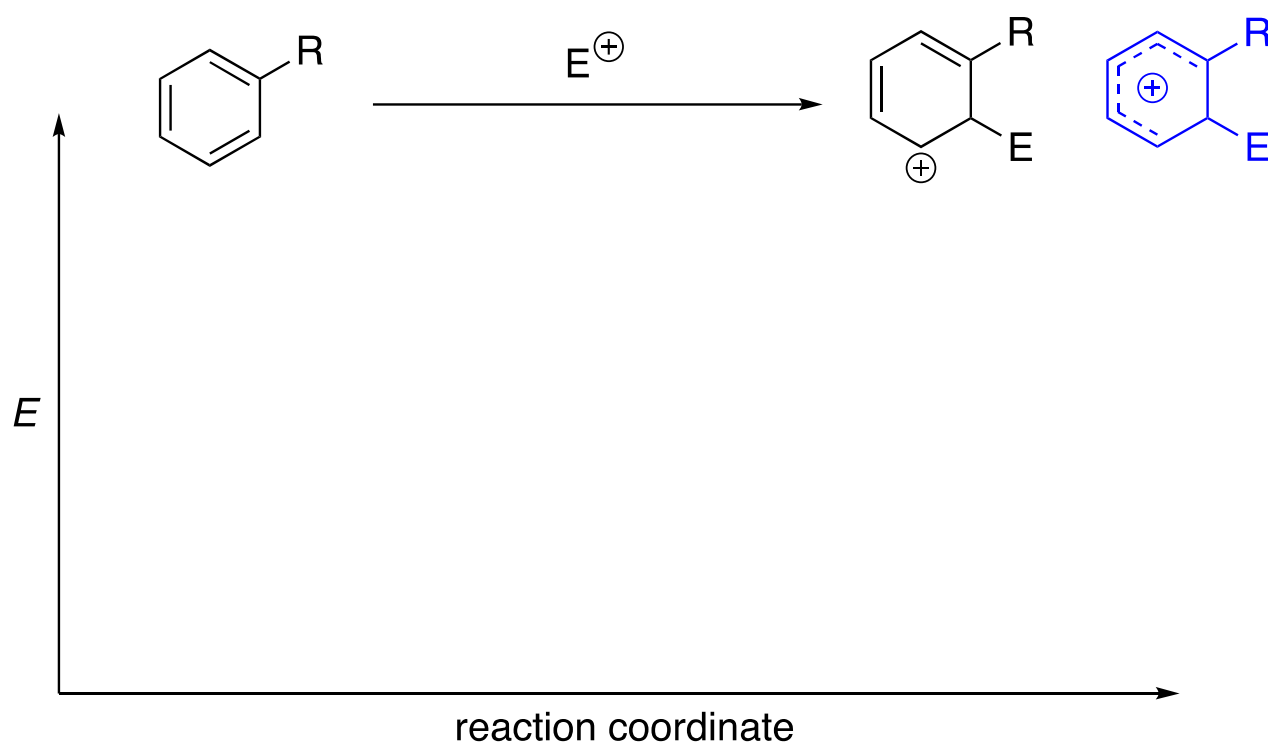
Clemmensen reduction



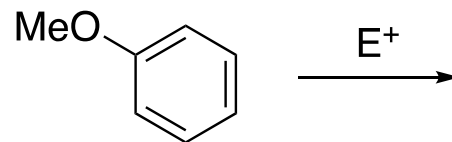
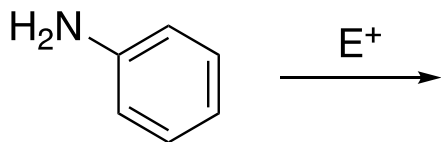
- In the rate-determining step of EAS reactions, the aromatic ring acts as a nucleophile
- **Electron-rich rings** react more **rapidly** than benzene in EAS reactions



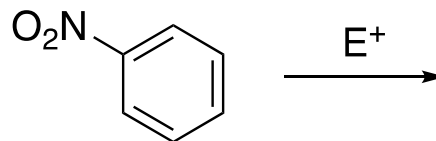
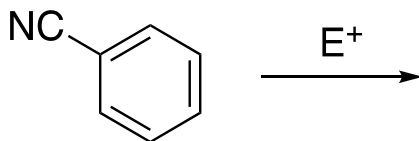
- In the rate-determining step of EAS reactions, the aromatic ring acts as a nucleophile
- **Electron-poor rings** react more **slowly** than benzene in EAS reactions



- Substituents linked to benzene rings “direct” electrophilic substitution to specific positions (*ortho*, *meta*, or *para*)
- $-\text{NR}_2$, $-\text{OR}$, and related groups are ***ortho/para* directors** (the para product generally dominates for steric reasons)

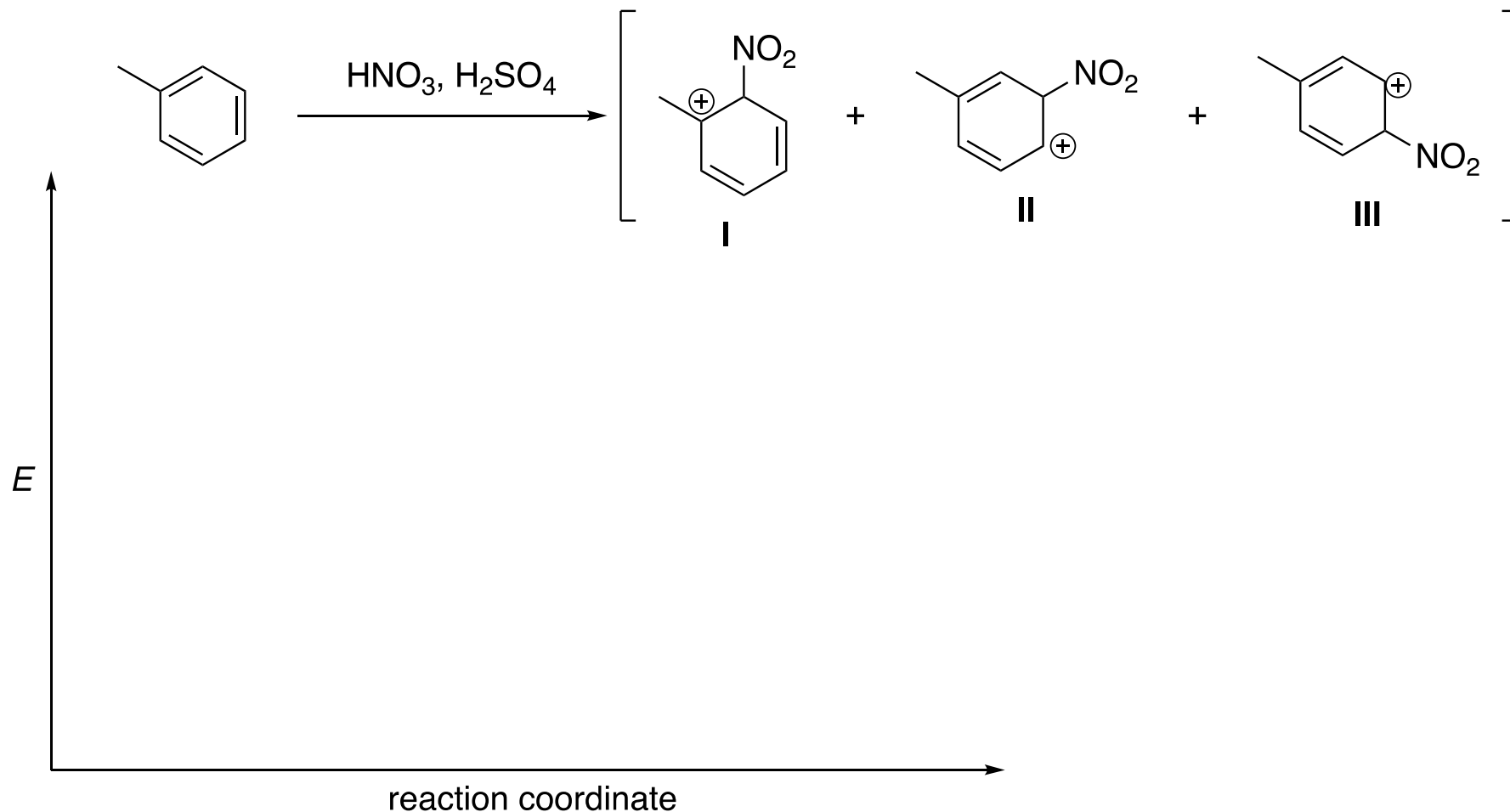


- $-\text{CN}$, $-\text{NO}_2$, and related groups are ***meta* directors**

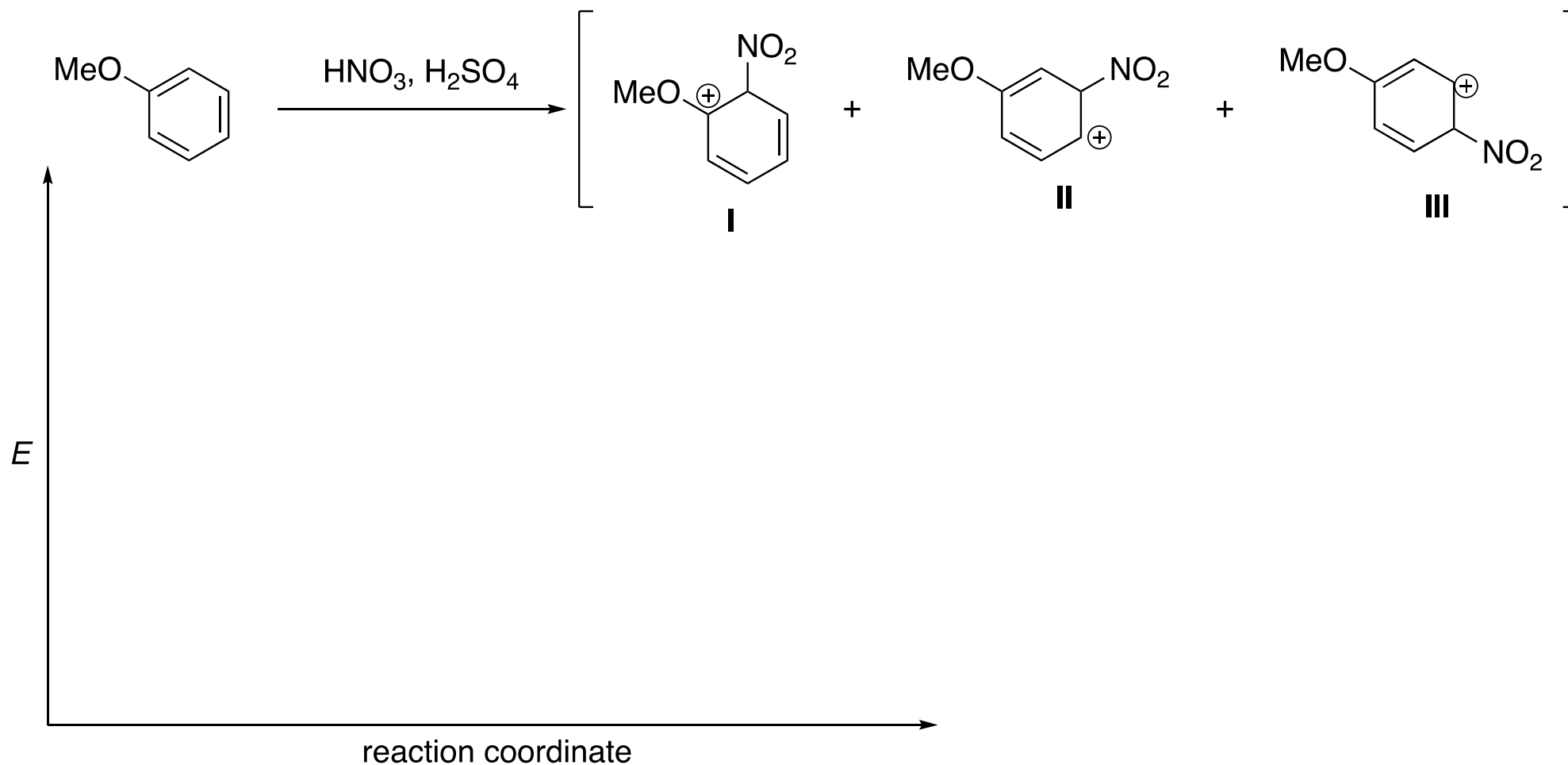


- Do you notice a pattern?

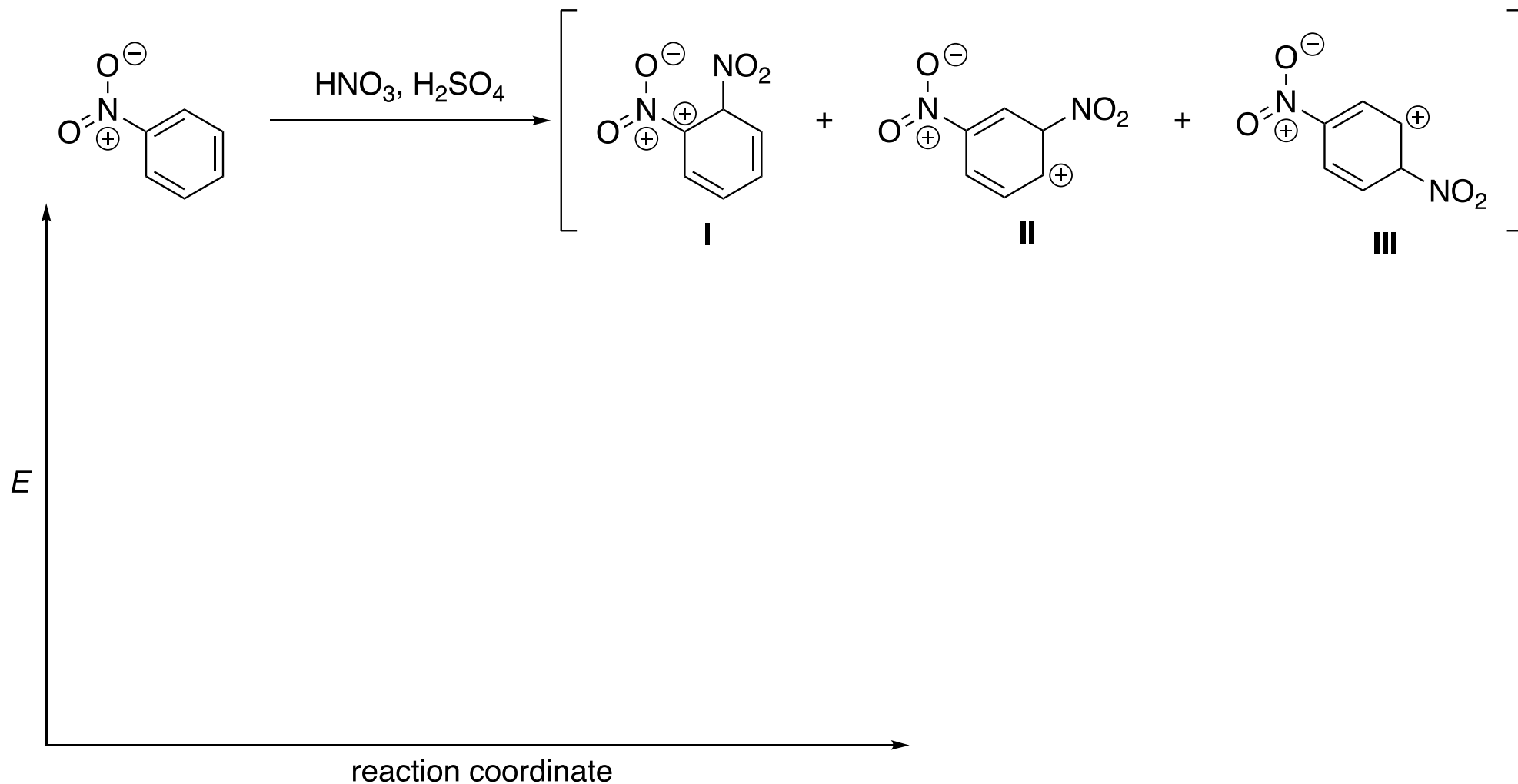
- Alkyl groups are also *ortho/para* directors and activate benzene toward electrophilic substitution (they are electron-donating groups)
- Consider the relative stabilities of the possible isomeric arenium ions **I** – **III** in the nitration of toluene



- Groups that are **electron donating by resonance** are *ortho/para* directors
- Consider the relative stabilities of the possible isomeric arenium ions **I** – **III** in the nitration of anisole

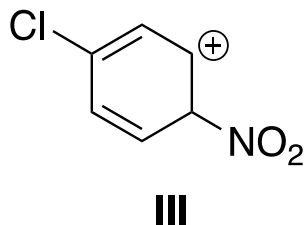
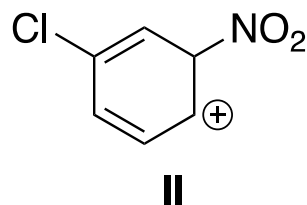
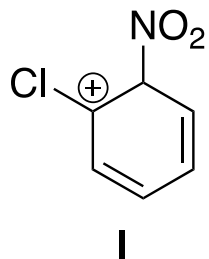


- Groups that are **electron withdrawing by resonance** are *meta* directors and deactivating toward EAS
- Consider the relative stabilities of the possible isomeric arenium ions **I** – **III** in the nitration of nitrobenzene

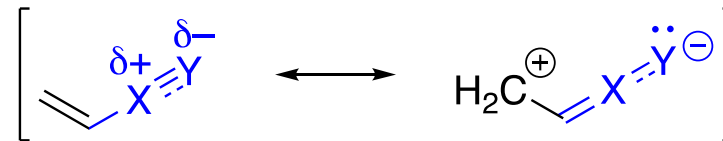
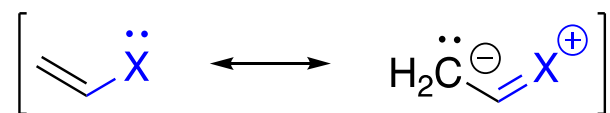


18.7 – 18.9 Halogens are ortho/para Directors but Deactivating Toward EAS

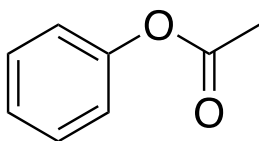
- The halogens fit the structural pattern of a **resonance electron-donating group**...
- ...but they are strongly **withdrawing by induction**, and so deactivate the ring toward reaction with electrophiles
- For regiochemical predictions, resonance wins the day (as usual)



- Review the general structures of electron-donating and -withdrawing groups



- Watch out for “competitive delocalization” of lone pairs, which moderates electron-donating strength (these groups are still *ortho/para* directing, though)



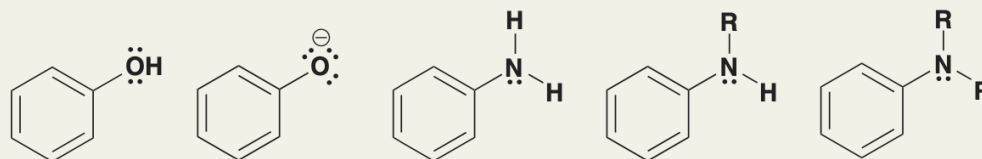
- Most resonance EWGs are moderately deactivating; however there are three important **strong deactivators**: $-\text{NR}_3^+$, $-\text{NO}_2$, and $-\text{CF}_3$

18.10 Determining the Directing Effects of a Substituent | 2 of 2

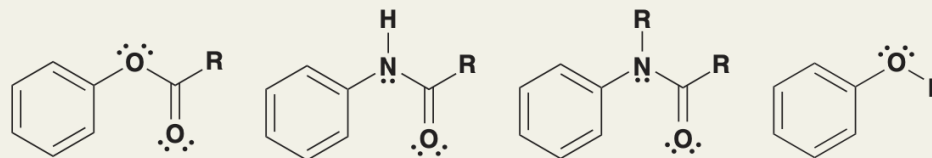
TABLE 18.1 A LIST OF ACTIVATORS AND DEACTIVATORS BY CATEGORY

Activators

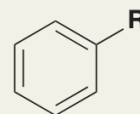
Strong



Moderate



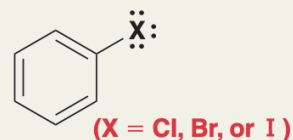
Weak



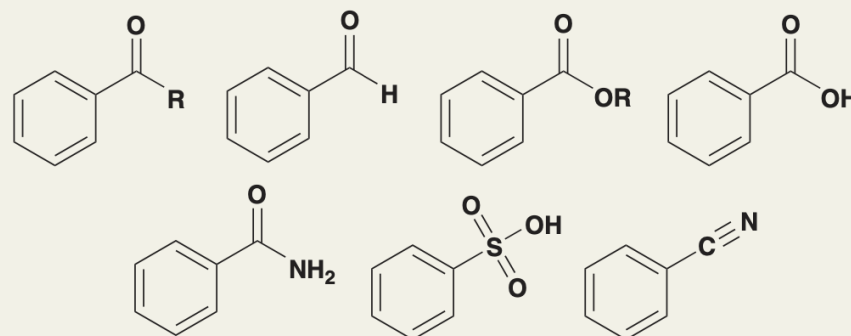
ortho-para
directors

Deactivators

Weak

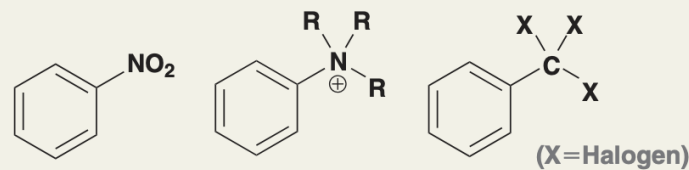


Moderate



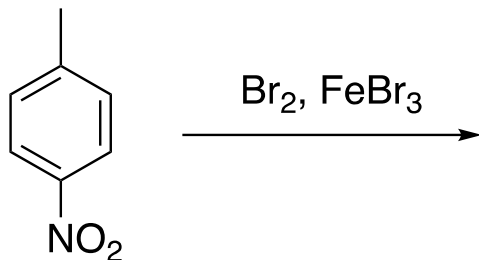
meta
directors

Strong

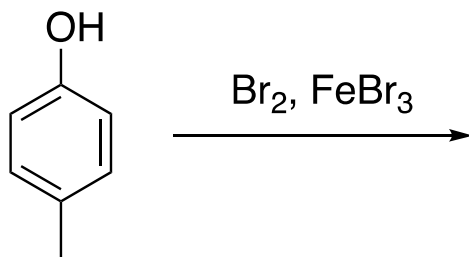


18.11 Disubstituted Benzenes and Beyond

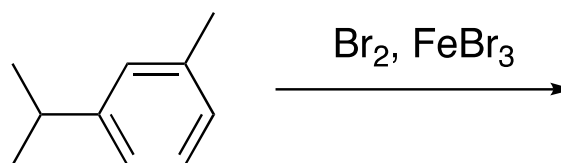
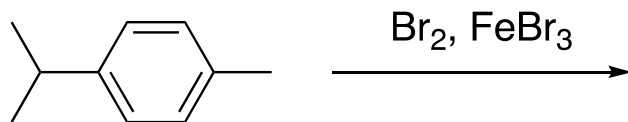
- When considering disubstituted benzenes, look for **reinforced positions**



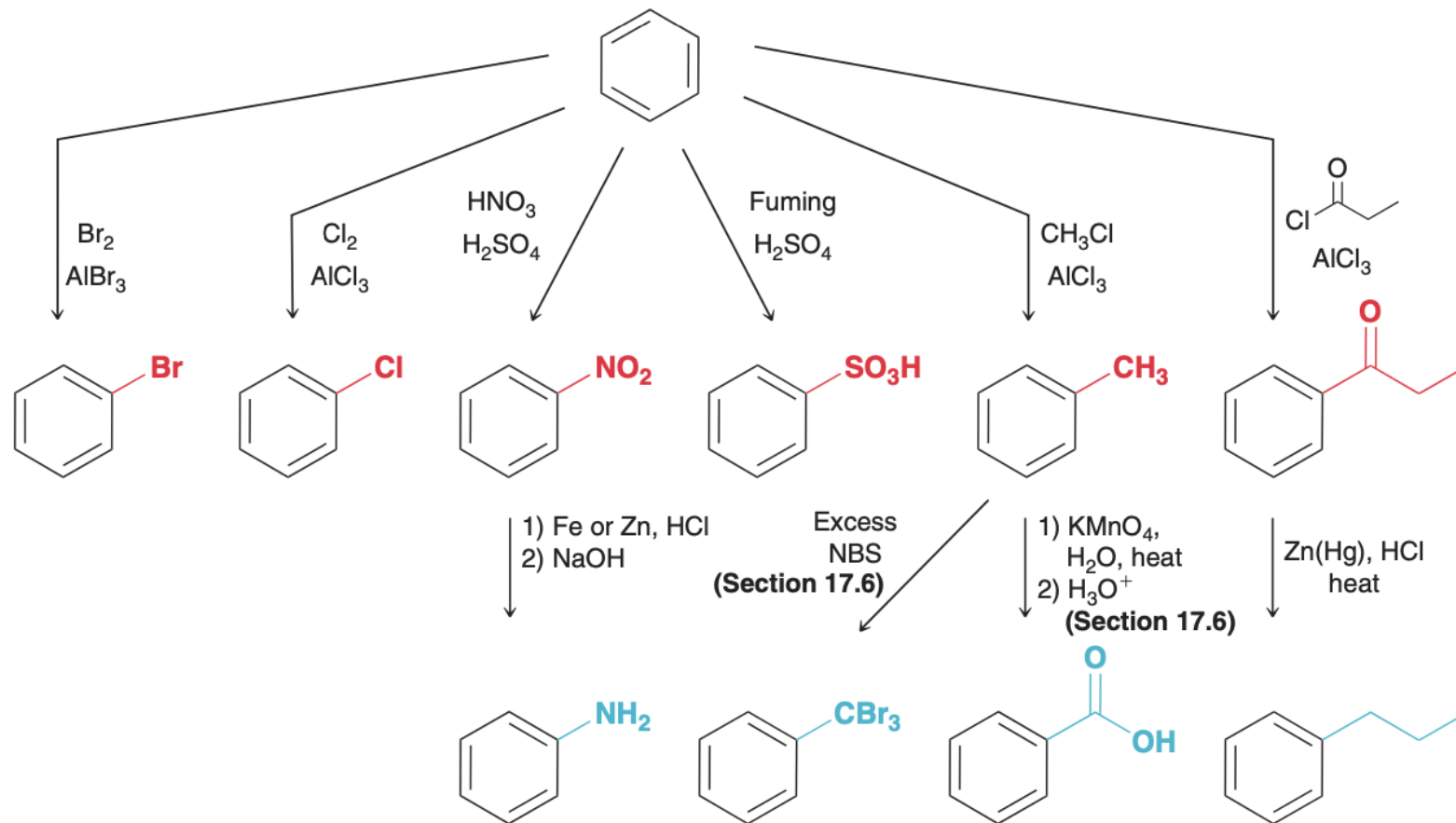
- When two substituents are competing for different positions, the stronger activator dictates the major product



- In reactions of 1,3- and 1,4-dialkylbenzenes, the more sterically accessible positions react more rapidly



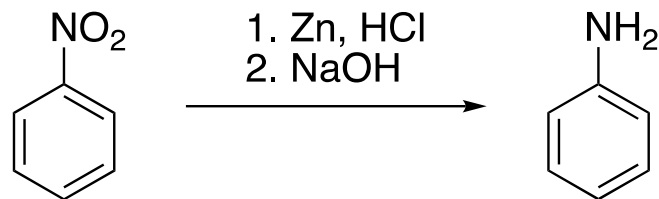
18.12 Synthetic Strategies for Monosubstituted Benzenes



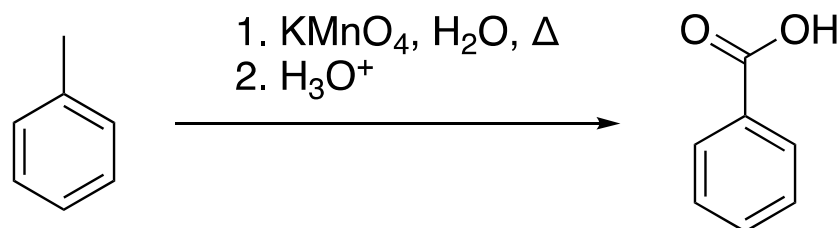
18.12 Functional Group Interchange Reactions that Change Directing Effects

- Synthesis of disubstituted benzenes may require functional group interchange (FGI)

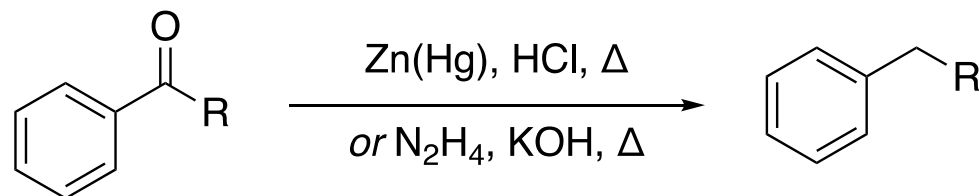
- Reduction of the nitro group to the amino group



- Oxidation of alkylbenzenes to benzoic acids

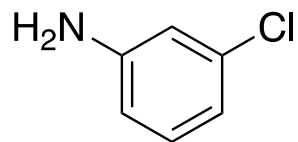


- Wolff-Kishner and Clemmensen reductions



18.12 Designing a Synthesis of a Disubstituted Benzene

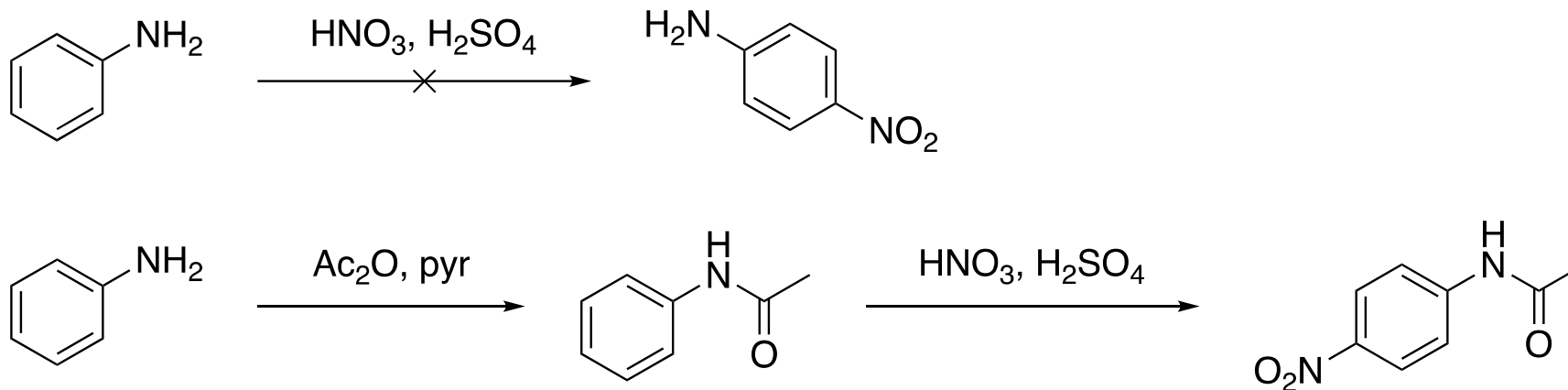
- How could we synthesize 3-chloroaniline from benzene?



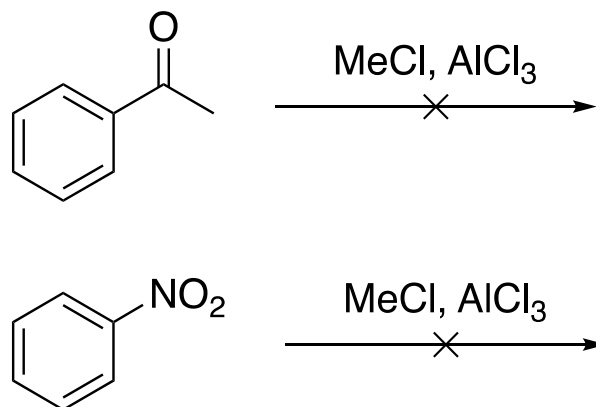
3-chloroaniline

18.12 Important Limitations of Electrophilic Aromatic Substitutions

- Anilines (ArNH_2) are incompatible with strongly Lewis acidic and Brønsted acidic conditions; they must be converted to amides first via reaction with acetic anhydride and pyridine

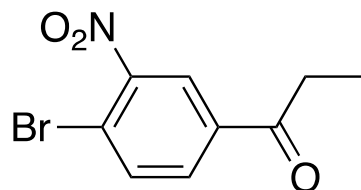


- Friedel-Crafts reactions require an activated or only weakly deactivated (halogens) starting material

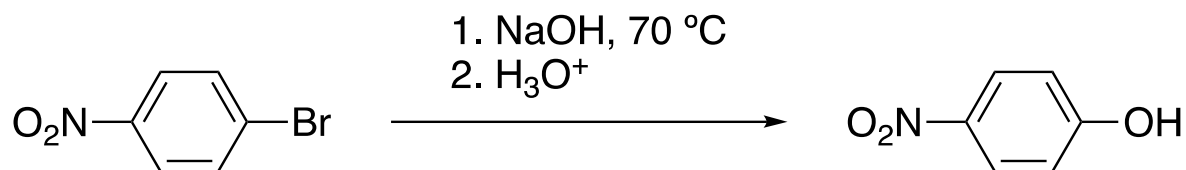


18.12 Retrosynthetic Analysis of Polysubstituted Benzenes

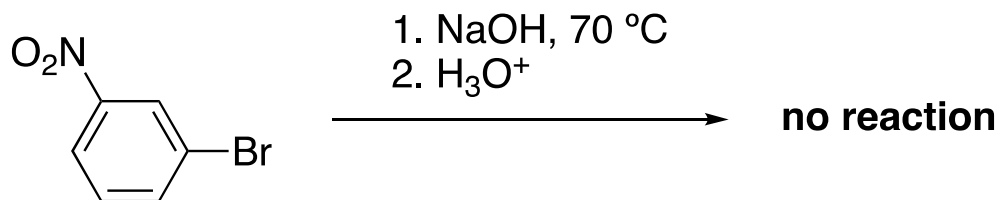
- How could we synthesize the trisubstituted benzene below from benzene? Use retrosynthetic analysis!



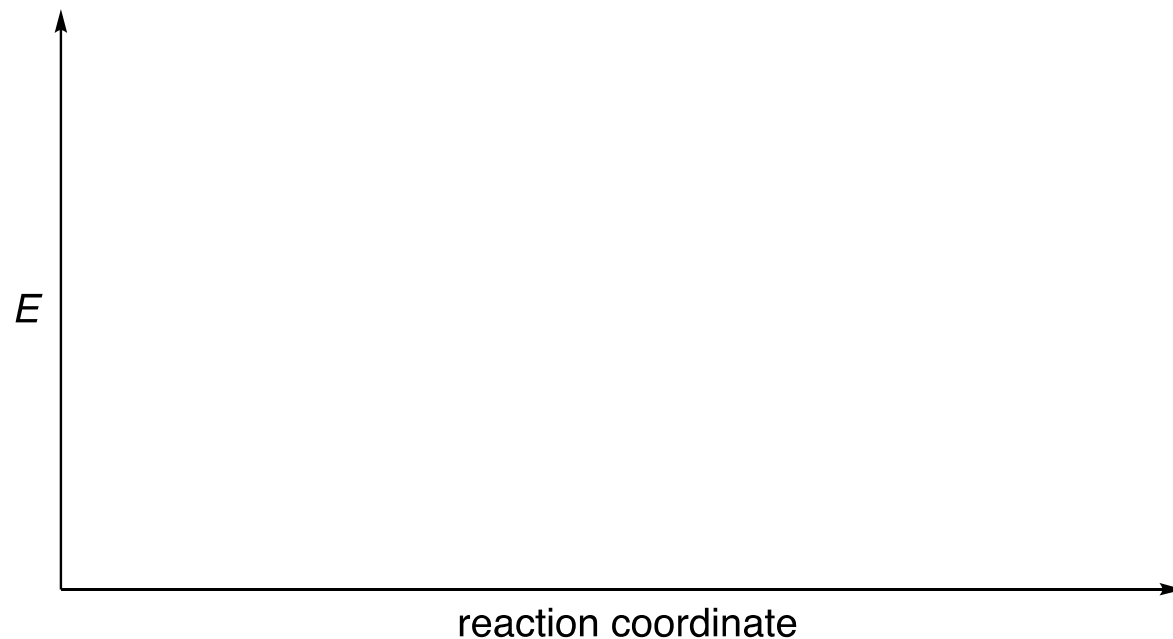
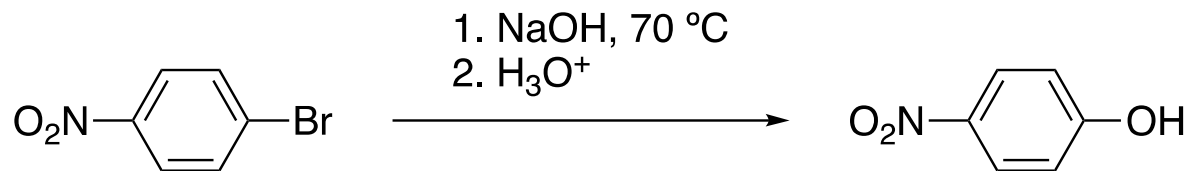
- In a **nucleophilic aromatic substitution (NAS)** reaction, a nucleophile displaces a good leaving group linked to an aromatic or heteroaromatic ring



- Three conditions for NAS reactions of benzenes:
 - The ring must bear a powerful electron-withdrawing group (typically nitro)
 - The ring must bear a good leaving group (typically halide)
 - The leaving group must be *ortho* or *para* to the electron-withdrawing group

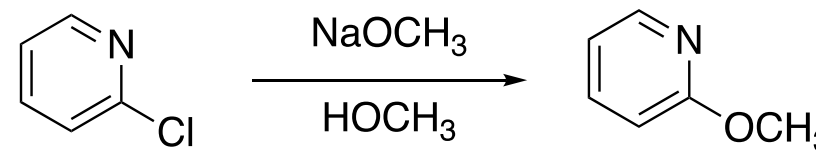
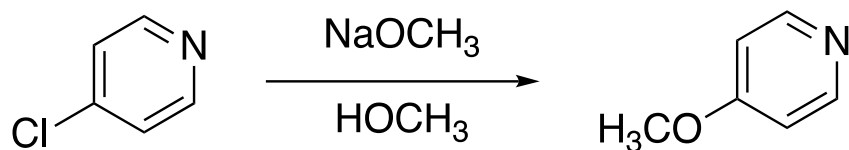


- The canonical mechanism involves rate-determining nucleophilic addition followed by beta-elimination (**S_NAr**)
 - The nucleophile adds to the carbon bearing the leaving group (Ad_{N})
 - The leaving group departs (E_{β})

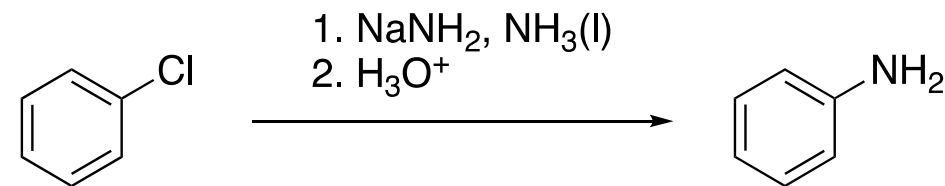
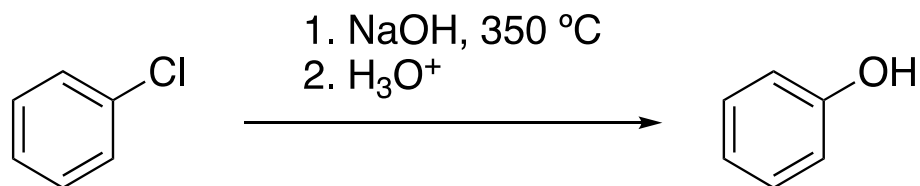


- An *ortho* or *para* EWG is needed for stabilization of the intermediate **Meisenheimer complex**

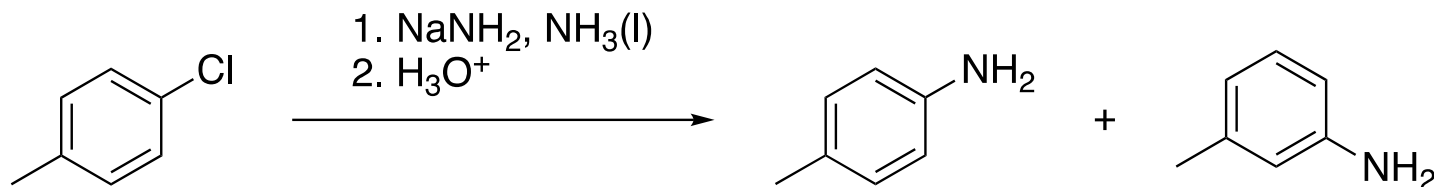
- Nucleophilic aromatic substitutions of pyridines and other electron-deficient heterocycles have great practical importance in the pharmaceutical industry
- For example, 2- and 4-halopyridines (*not* 3-halopyridines!) react rapidly with alkoxides at room temperature



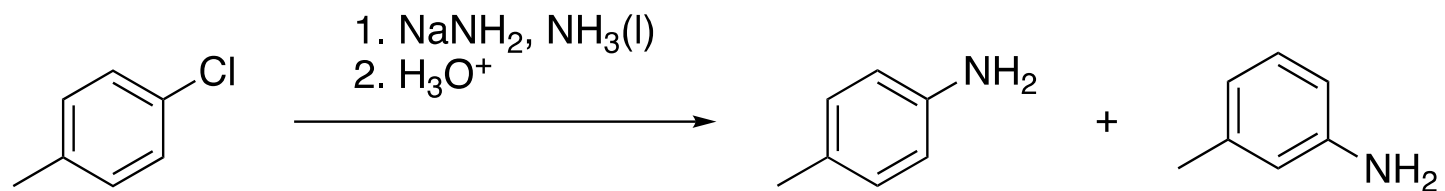
- Simple halobenzenes react with hydroxide at elevated temperature (350 °C) and amide anion in nucleophilic aromatic substitutions



- When another substituent is present, the regiochemical outcome is strange...

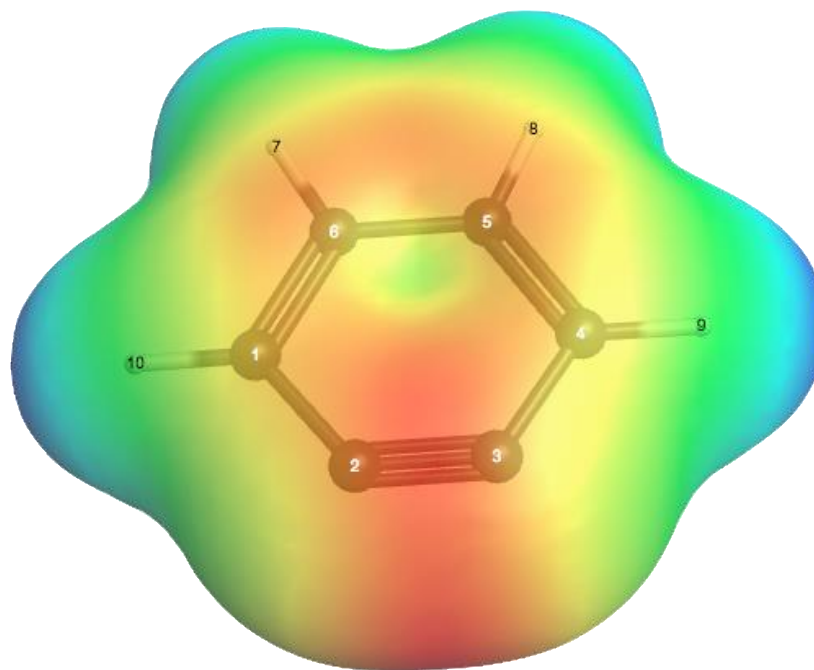
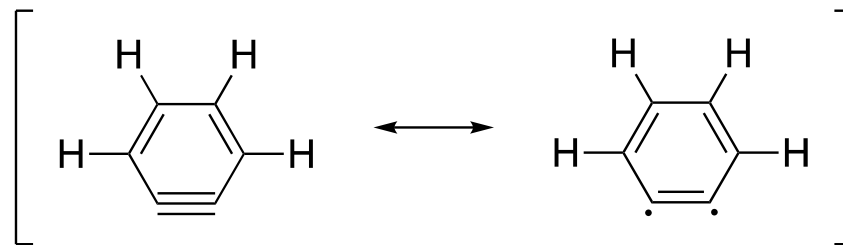


- An **elimination-addition** mechanism involving a **benzyne** intermediate accounts for both products



- Benzynes can be used as Diels-Alder dienophiles
- Arynes** derived from other aromatic compounds such as pyridine have also been observed

- The triple bond in benzyne is not linear and the carbons are not quite *sp* hybridized
- The pi bond in the plane of the ring is “bent” and has diradical character



18.15 Determining the Mechanism of an Aromatic Substitution Reaction

- Three mechanisms of aromatic substitution: electrophilic aromatic substitution, S_NAr , and elimination-addition
- *The intermediate*: arenium ions in EAS, Meisenheimer complexes in S_NAr , and benzyne in elimination-addition
- *The leaving group*: In EAS the leaving group is the electrofuge H^+ ; in NAS the leaving group is the nucleofuge X^-
- *Substituent effects*: electron-withdrawing groups slow down electrophilic substitution but accelerate nucleophilic substitution

